AI/ML
Tools of the trade

Thomas Britton

Who am I? / Who am I not?

- PhD in experimental high energy physics
  - Syracuse 2016 LHCb
- Postdoc with GlueX at Jefferson lab
- Staff scientist with Jefferson Lab
- Have developed several projects which utilize AI/ML
  - With an aim of deployment

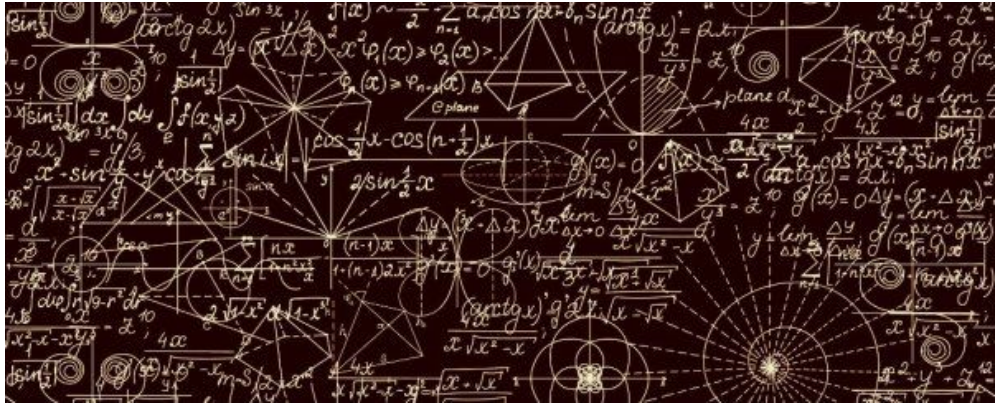
- **NOT** a formally trained data scientist

So You Want to Take Over the World…

…Or You Have Some Problems….

The Problems Are:

- **Too complex**
  - (highly dimensional)



- **Not well modeled**
  - No analytic solution

- **Not feasible**
  - Too much data. Not enough expertise/people

Or You Are Just Too Lazy….

Seriously, maybe the problem is too repetitive and you just can’t be bothered

*probably contains modifications on a repetitive theme

Introducing AI/ML!

"AI (Artificial Intelligence) and ML (Machine Learning) are two related fields that are revolutionizing many industries, including physics. AI refers to the development of computer systems that can perform tasks that typically require human intelligence, such as understanding natural language, recognizing images, and making decisions.

ML is a subfield of AI that focuses on developing algorithms that can learn from data and make predictions or decisions based on that learning. ML algorithms are designed to automatically improve their performance over time as they are fed more data.

So why should graduate students in physics try AI/ML? Well, there are several reasons. First, AI/ML has already proven to be incredibly useful in physics research. It can be used to analyze large datasets, model complex systems, and even discover new physical principles. Second, learning AI/ML will give students valuable skills that are in high demand in many industries, including finance, healthcare, and tech. Finally, exploring AI/ML can be an intellectually stimulating and rewarding experience that opens up new avenues of research and creativity.

Overall, AI/ML is a rapidly growing field that offers exciting opportunities for physics research and career development."

Thanks ChatGPT!

Where to begin?

Okay ChatGPT, what should these students learn first?

Before Beginning, an Overview

"One piece of advice for someone just starting to learn AI/ML is to focus on building a solid foundation in the fundamentals before diving into advanced topics.

AI/ML is a complex and rapidly evolving field with many different algorithms, techniques, and frameworks to choose from. It can be overwhelming to try to learn everything at once. Instead, start by focusing on the basics: linear algebra, calculus, probability theory, and statistics. These are the mathematical foundations of AI/ML and will be essential for understanding how algorithms work and how to apply them effectively.

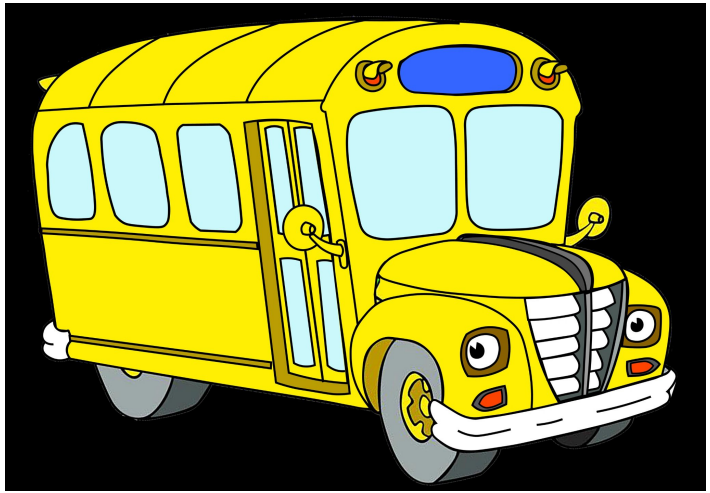
Once you have a good grasp of the fundamentals, you can start exploring different AI/ML techniques and frameworks. Don't be afraid to experiment with different algorithms and datasets to gain practical experience. And remember, AI/ML is a team sport - collaborate with others, ask questions, and seek feedback from experts in the field.

Finally, be patient and persistent. AI/ML can be challenging, but with dedication and hard work, you can achieve your goals and make significant contributions to the field."

Yes, ChatGPT that is all true. But it doesn't sound like as much fun....

Take Chances, Make Mistakes, and Get Messy!

Yes, Ms. Frizzle that does sound like more fun!



Disclaimer

AI/ML is a **large**, *complex*, rapidly changing field.

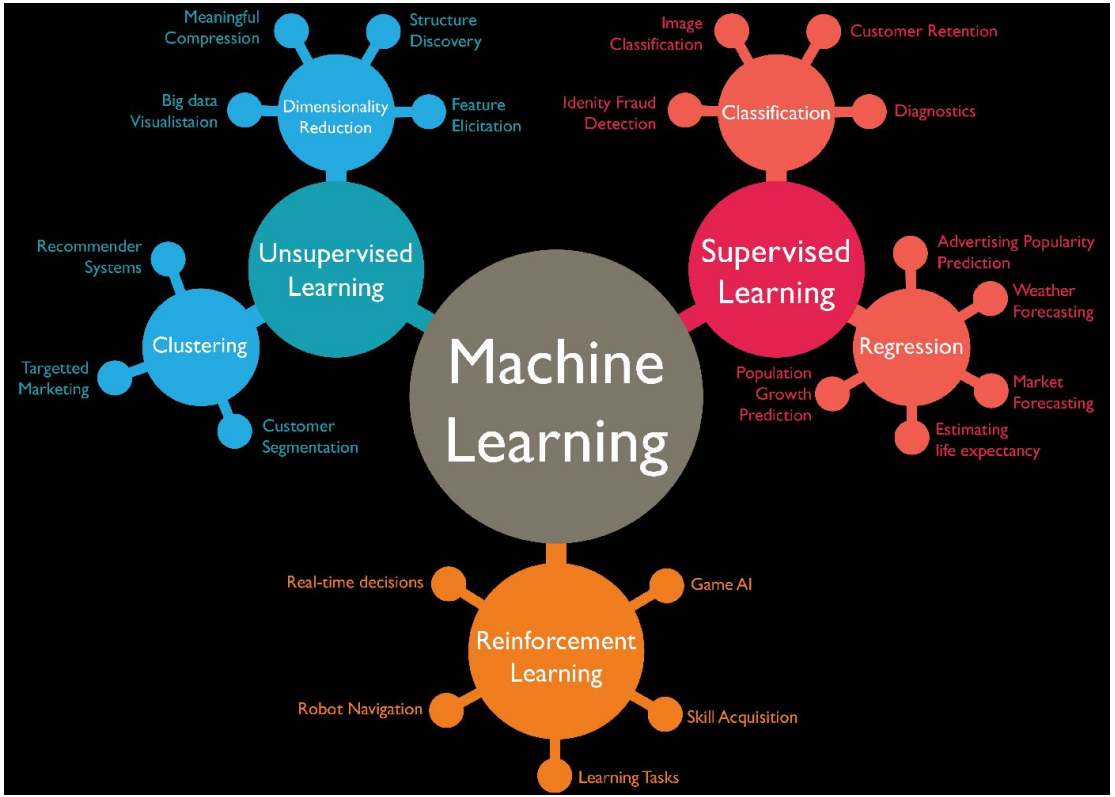
Much of what you find today may be obsolete in a matter of months. Four hours is not nearly enough time to get the basics let alone obtain any sort of mastery over the material.

These tutorials, instead, should be viewed as a dip in the shallow end of a new pool. I will try to present more foundational things and give a chance to explore these concepts in a more curated way.

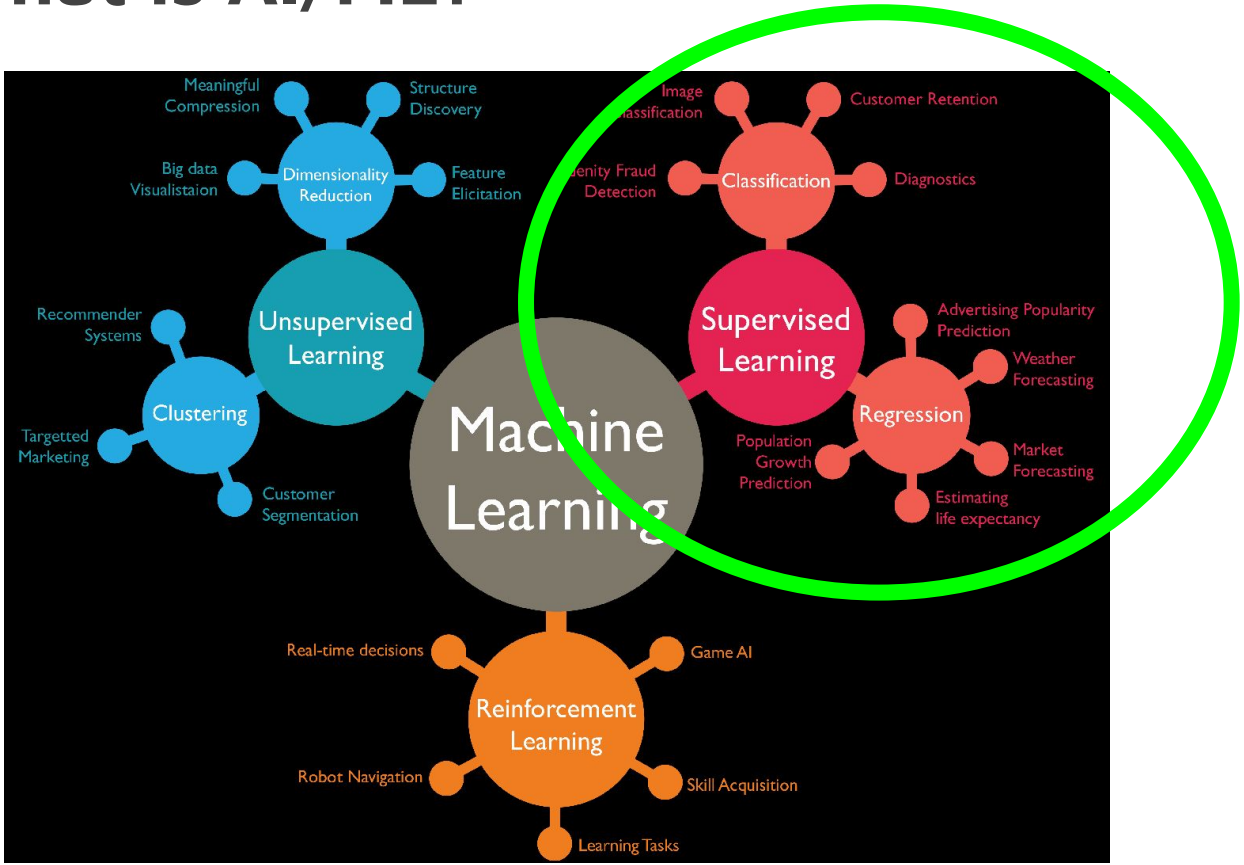
Rule 0: Be willing to be wrong

Rule 1: Know your data

What is AI/ML?


Machine Learning
Unsupervised Learning
Dimensionality Reduction
Meaningful Compression
Structure Discovery
Big data Visualistaion
Feature Elicitation
Clustering
Recommender Systems
Targetted Marketing
Customer Segmentation
Supervised Learning
Classification
Image Classification
Customer Retention
Idenity Fraud Detection
Diagnostics
Regression
Advertising Popularity Prediction
Weather Forecasting
Population Growth Prediction
Market Forecasting
Estimating life expectancy
Reinforcement Learning
Real-time decisions
Game AI
Robot Navigation
Skill Acquisition
Learning Tasks

What is AI/ML?


Meaningful Compression
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Recommender Systems
Clustering
Targetted Marketing
Customer Segmentation
Machine Learning
Image
Customer Retention
Classification
Diagnostics
Supervised Learning
Advertising Popularity Prediction
Weather Forecasting
Regression
Population Growth Prediction
Market Forecasting
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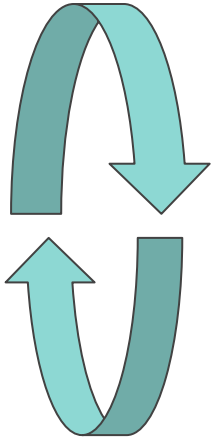
Learning Tasks

How Babies Tend to Learn

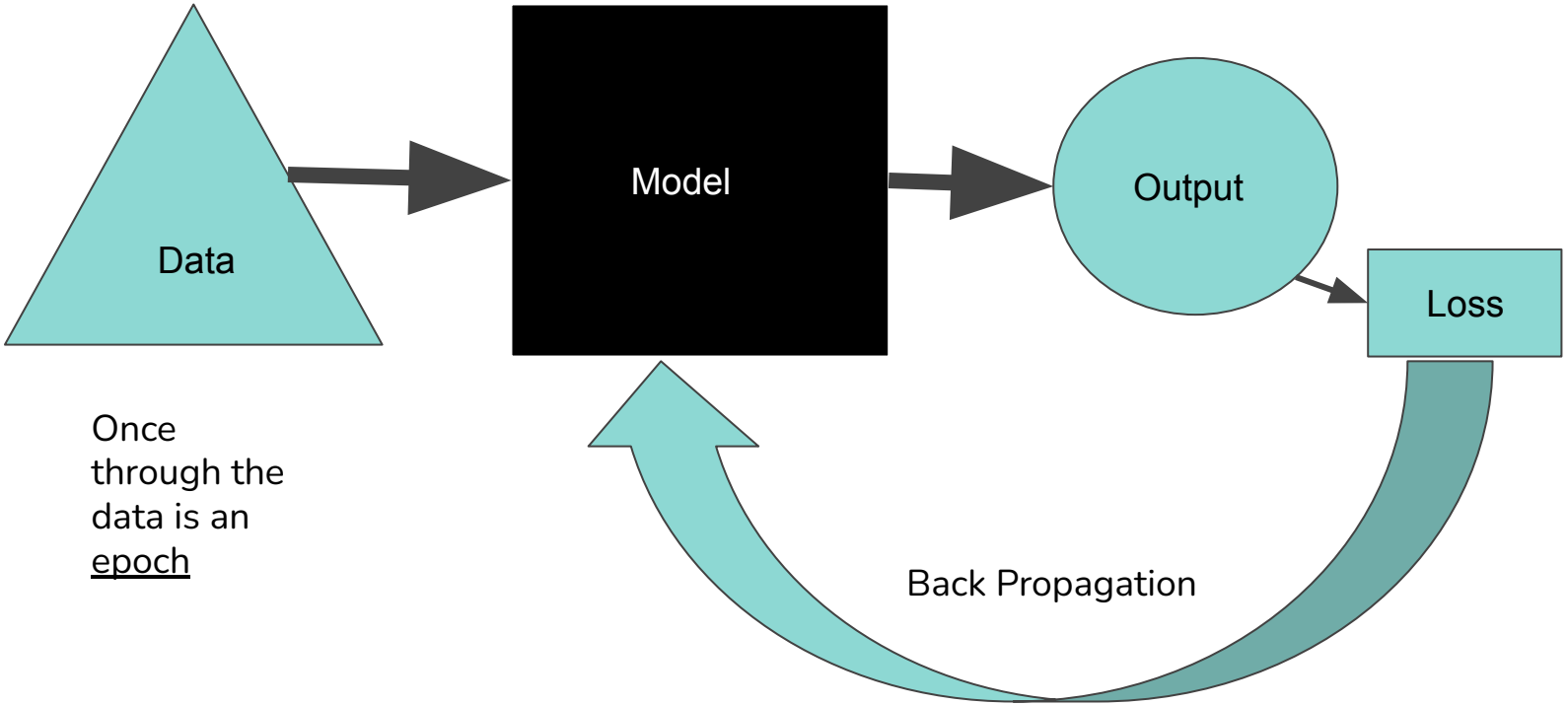
- Parent: *What's this?*
- Child: *Dog!*
- Parent: *Yes!*
- Parent: *What's this?*
- Child: *Dog!*
- Parent: *No, that's a cat*
- ....



*Each trip through the book is an <u>epoch</u>

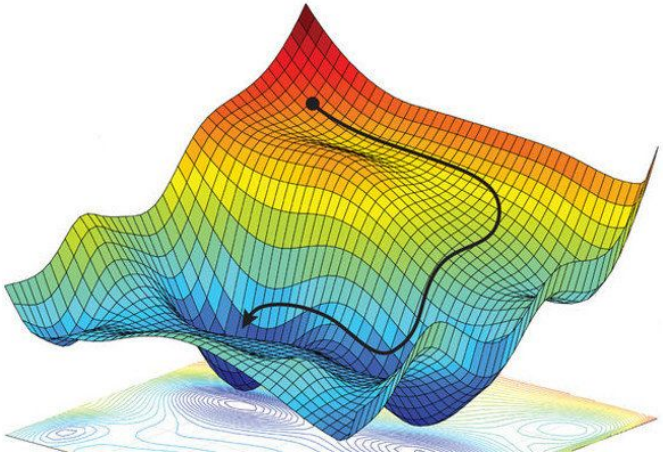
My mother reading to my son

The Major Parts of AI



Once through the data is an <u>epoch</u>

The Loss Function is Key



The loss function is the mathematical way of telling an AI what to learn. Wrong loss function can have catastrophic outcomes.

If your metric of success is a lack of cancer progression after 3 months then terminating the patient would certainly fit the bill.

This would be an example of a poor loss function.

Know Your Data

Before you try anything you need to explore your data.

Representation of the data is dependent on the goals, the techniques, the data.

For example, if relative geometric location matters you might want to keep it as an image. Maybe it is sparse and linearizing/dropping the empty space makes more sense.

Seriously Know Your Data

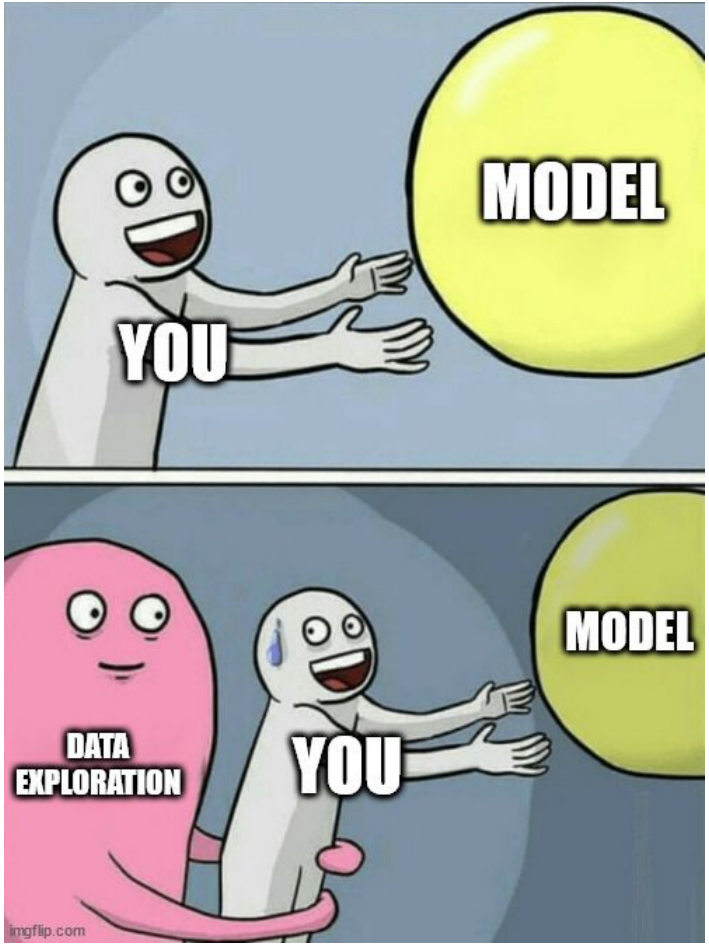
rush to model construction

explore and understand your data first

One Last Thing

Know your data

[An example]



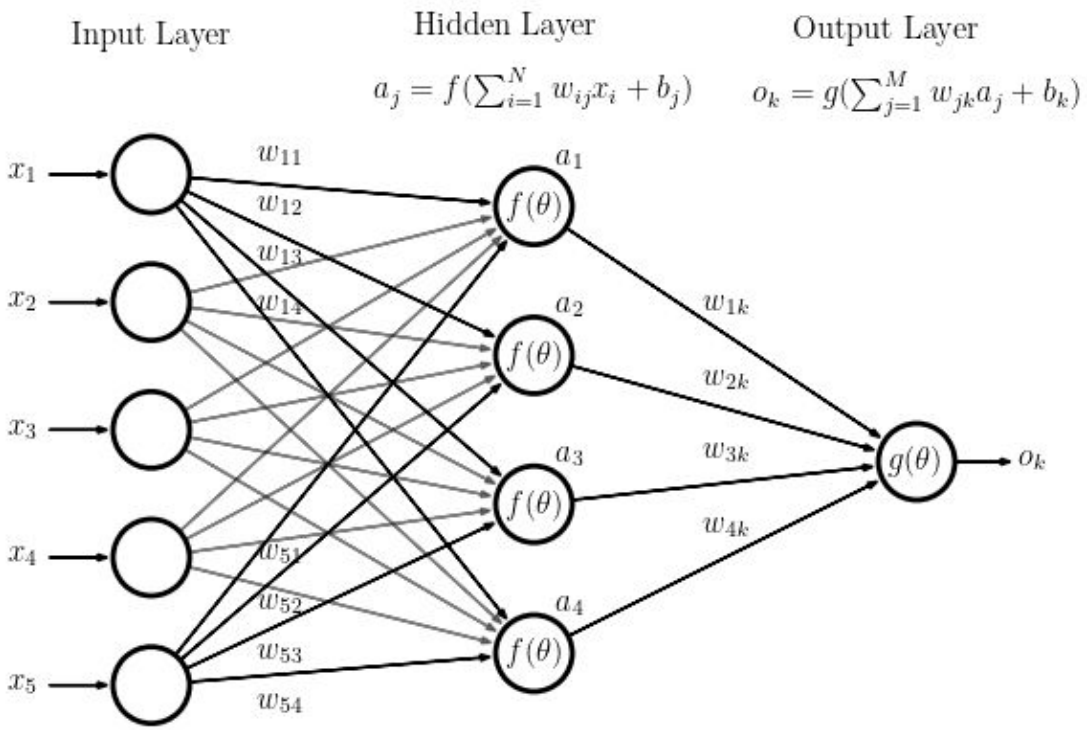
Before Beginning, an Overview

The goal is to give everyone a cursory overview with some hands on practice. This first tutorial will be introducing everyone to the tools and some underpinnings of AI/ML

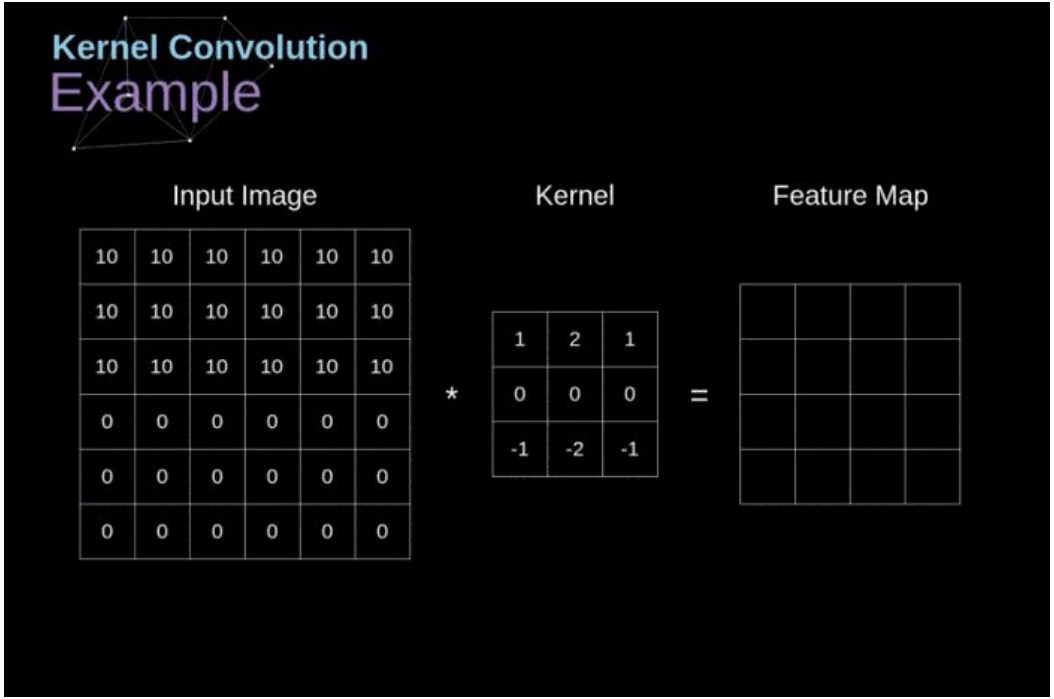
TLDR: We'll be using google collab

But feel free to use any other system...they all roughly have feature parity, certainly for what we are going to do whatever works.

The Basic Neural Net Node

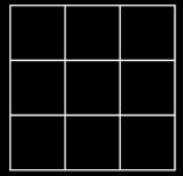


Convolutions



Edge Detection

Using Kernel Convolution



MNIST

Resources

[3 brown 1 blue]

[O'Reilly books]

Those that came before